

AISHI TIWARI

☎ +91-9354414404 ✉ aishitiwari1@gmail.com  LinkedIn

🔗 LeetCode Portfolio > Github

EDUCATION

Maharaja Agrasen Institute of Technology, GGSIPU

2024 – 2028

B.Tech - Computer Science and Technology - CGPA: 9.10/10

Delhi

Sardar Patel Vidyalaya (Class XII)

Completed 2024

CBSE, Science Stream (PCM) - SCORE: 90.0

Delhi, India

TECHNICAL SKILLS

Machine Learning and AI: PyTorch, Scikit-learn, Keras, Hugging Face, YOLO, Roboflow, NumPy, Matplotlib, CUDA, LangChain

System and Tools: Linux, NASM, Git, GitHub, VS Code, Jupyter, Conda, Bash, REST APIs, Postman

Programming Languages: Python, C++, C, Java, SQL, JavaScript, TypeScript

Web Deployment: Flask, FastAPI, React, Node.js

Data Structures and Problem Solving: Solved 350+ problems on LeetCode along with regular Contest participation.

EXPERIENCE

Indian Institute of Technology with Larsen and Toubro, Delhi

April 2026 – ongoing

Machine Learning & Systems Intern

On-Site

- Designed a multimodal dynamic inference system that selects the most suitable model at runtime using semantic image embeddings, achieving 94% routing accuracy to the best model across 5+ expert models.
- Built pipelines combining VLMs, LLMs, and embedding-based retrieval for expert profiling and runtime best model selection.
- Co-authored "VLM-Guided Expert Routing for Domain-Robust Object Detection," submitted to AAAI 2027.

NITI Aayog, Government of India

July 2025 – Sept 2025

Science and Technology Intern

On-Site

- Built an end-to-end ML pipeline for processing high-resolution satellite imagery and automatically mapping infrastructure across 112 aspirational districts of India.
- Developed and trained a YOLO-based object detection system using self-annotated datasets created with Roboflow.
- Processed and annotated 500+ satellite images of hospitals and health centres for infrastructure profiling and automated analysis, contributing directly to Data based policy making

PROJECTS

Helium – Compiler for a Custom Programming Language

July 2026 | GitHub Link

- Developing **Helium**, a compiler in C++ that generates x86-64 Linux executables using NASM and LD.
- Designed a modular compiler architecture with a hand-written recursive-descent parser, cleanly separating lexical analysis, parsing, and code generation.
- Implemented 10+ core language features, including conditionals, functions, return statements, and I/O, with correct assembly-level control flow without relying on compiler frameworks or backend libraries.

Customer Support Intelligence System

April 2026 | GitHub Link

- Developed a **plug-and-play AI middleware** that predicts optimal human handoff points in support bot conversations with 89% escalation accuracy on simulated user conversations.
- NLP features include sentiment & repetition analysis and fallback detection from a synthetic dataset of 400+ conversations.
- Deployed a Flask API for real-time support recommendations, reducing decision latency by 2–3 seconds compared with LLM-based approaches and cutting cost by 40%

myRepoMate – Personalized GitHub Repository Onboarding

May–June 2026 | GitHub Link

- Developed a full-stack application that analyzes GitHub repositories and generates personalized onboarding guides for developers.
- Designed a dynamic multi-key API rotation mechanism that automatically detects rate limits and failed keys, maximizing throughput through seamless failover.
- Deployed the application to 20+ developers and iteratively improved the product based on user feedback.

ACHIEVEMENTS & CERTIFICATIONS

- 5X Hackathon finalist +2X winner at inter and intra-college levels+ Finalist Smart India Hackathon 2025
- **SheFi – 2025 Cohort Scholar:** Selected among top students in India for a highly competitive fintech, and innovation program.
- Among the **top 1,500 programmers in Google Big Code**, a competitive nationwide coding challenge.
- Among the top 8% Adobe University Hackathon.
- Certifications: Andrew NG's Machine Learning Specialization (Coursera, DeepLearningAI & Stanford Online), Machine Learning Learning Plan (AWS) Advanced MCP+ Introduction to MCP (Anthropic).
- Appointed Technology Head of TechXtract- The Official Society of the Computer Science Department.

EXTRACURRICULARS

- 4x Elocution/Debate Competition winner at inter-college and School level.
- Member, Student Executive Sardar Patel Vidyalaya - Selected among the top 15 students for the student board overseeing event management and student communications.